

Technical Document #225: Natural Language Processing - Comprehensive Overview

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Machine translation converts text from one language to another. Neural machine translation using encoder-decoder architectures has achieved human-level performance.

Word embeddings represent words as dense vectors in continuous space. Word2Vec, GloVe, and FastText are popular word embedding techniques.

Named entity recognition (NER) identifies and classifies named entities in text into predefined categories such as person names, organizations, and locations.

Tokenization is the process of breaking text into individual units called tokens. Subword tokenization methods like BPE handle out-of-vocabulary words gracefully.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Key Takeaways:

- Question answering systems extract answers from a given context.
- Machine translation converts text from one language to another.
- Named entity recognition (NER) identifies and classifies named entities in text into predefined categories such as person names, organizations, and locations.